

Student's Name:.....Signature:

525/1
CHEMISTRY
Paper 1
2025
 $2\frac{1}{2}$ Hours

July/August, 2025

CHEMISTRY-CONNECT

S5 END OF TERM 2 ASSESSMENT

Uganda Advanced Certificate of Education

CHEMISTRY

Paper 1

2 Hours and 30 Minutes

INSTRUCTIONS TO STUDENTS:

This paper consists of two sections A and B. All items in both sections are compulsory.

*Answers to section A **must** be written in the spaces provided while for section B, the answers **must** be written on the answer sheets provided.*

SECTION A

(Items in this section are all compulsory)

Item 1: During a chemistry lesson on gaseous reactions, learners were asked to investigate an unknown gaseous hydrocarbon, X. Sharon conducted an experiment where she exploded 20 cm³ of hydrocarbon X with 100 cm³ of excess oxygen. After cooling, the total gas volume was 90 cm³, and after passing the mixture through aqueous potassium hydroxide, the volume reduced to 50 cm³.

Hydrocarbon X burns in oxygen according to the following equation:-



Sharon is unsure how to use these results to identify hydrocarbon X. She presents her findings to the class for interpretation.

TASKS: As a chemistry student, Help Sharon to;

a) Understand what caused the drop in volume after explosion, and why did it reduce further after treatment with potassium hydroxide?

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b). Calculate the volumes of carbon dioxide and oxygen that reacted.

c). Deduce the molecular formula and molecular mass of hydrocarbon X.

d) Help Sharon to understand the impact of the gaseous product produced and how it can be mitigated.

Item 2: A group of forensic chemists are investigating organic substances they found at certain suspected illegal chemical dump. analysis gave them the molecular formula of the two compounds as M(C_3H_6) and N (CH_4). To gather more information about the compounds they also need to know the nature of the two compounds and their chemical reactivity. They are aware that you have some knowledge on organic chemistry and have come to you for advice.

Task: As a chemistry student help them to

i) Write the structural formulae and state the functional groups present in the organic compounds.

ii) Predict the reaction and write the mechanisms for the reaction of;

a) M with hydrogen gas

b) N with chlorine in presence of UV light

c) N with concentrated nitric acid at $(150-400)^\circ\text{C}$

(iii) Help the students to analyze the impacts of one of the compounds to the environment.

SECTION B

(All items in this section are all compulsory)

Item 3: During a classroom discussion on energy changes in chemical reactions, the teacher asked learners to suggest indirect ways of determining enthalpy changes for reactions that cannot be measured directly in the laboratory. One learner, Joel, proposed using Hess's Law.

To apply this, the class was asked to determine the standard enthalpy of formation (ΔH_f°) of methane gas (CH_4). They are provided with the following standard enthalpy changes of combustion:

$$\Delta H_c^\circ[\text{CH}_4] = -890 \text{ KJ/mol}$$

$$\Delta H_c^\circ[\text{H}_2] = -286 \text{ KJ/mol}$$

$$\Delta H_c^\circ[\text{C}_{(\text{Graphite})}] = -393 \text{ KJ/mol}$$

However, some learners are confused about how these combustion reactions can be used to determine the enthalpy of formation of methane using Hess's Law.

Task: As a chemistry student, help the learners to:-

- Understand the meaning of enthalpy of formation
- Using Hess's Law, construct an energy cycle or equation arrangement that links the combustion reactions to the formation reaction of methane:
- Use the provided data to calculate the standard enthalpy of formation (ΔH_f°) of methane.

d). Comment on the sign of the enthalpy of formation value obtained. What does it indicate about the stability of methane compared to its elements?

Item 4:

A chemistry student is studying the properties and reactions of the Group 2 elements (alkaline earth metals). The student is particularly interested in understanding the trends in these elements.

As part of the student's research, they come across a company that uses magnesium (Mg) in the production of fireworks and flares. The company is considering replacing Mg with another Group 2 element, calcium (Ca), to reduce costs.

However, the student is concerned that using Ca may not produce the same bright light and high temperature as Mg. The student decides to investigate the reactions of Mg and Ca with water, oxygen, acids, and halogens to determine which element is more suitable for the company's needs.

TASK: As a chemistry student, who has studied about group 2 elements,

(a) Explain the trends in atomic radius, electronegativity, and first ionization energy of the Group 2 elements. How do these trends affect the reactivity of these elements?

(b) Write and explain the reactions of Mg and Ca with:

- ✓ Water
- ✓ Oxygen
- ✓ Dilute hydrochloric acid (HCl)
- ✓ Chlorine (Cl₂)

Which element is more suitable for the company's needs?

(c) Discuss the general uses of the Group 2 elements in various industries

****END****

“Enjoy the Beauty of Chemistry”